Netflix Stock-Price Simulation Using the Heston Model

10-business-day Monte Carlo scenario analysis

Writer: Kan Wai Kit | Analysis period: 20 October 2026 to 3 November 2026 | 50,000 simulations

Executive summary

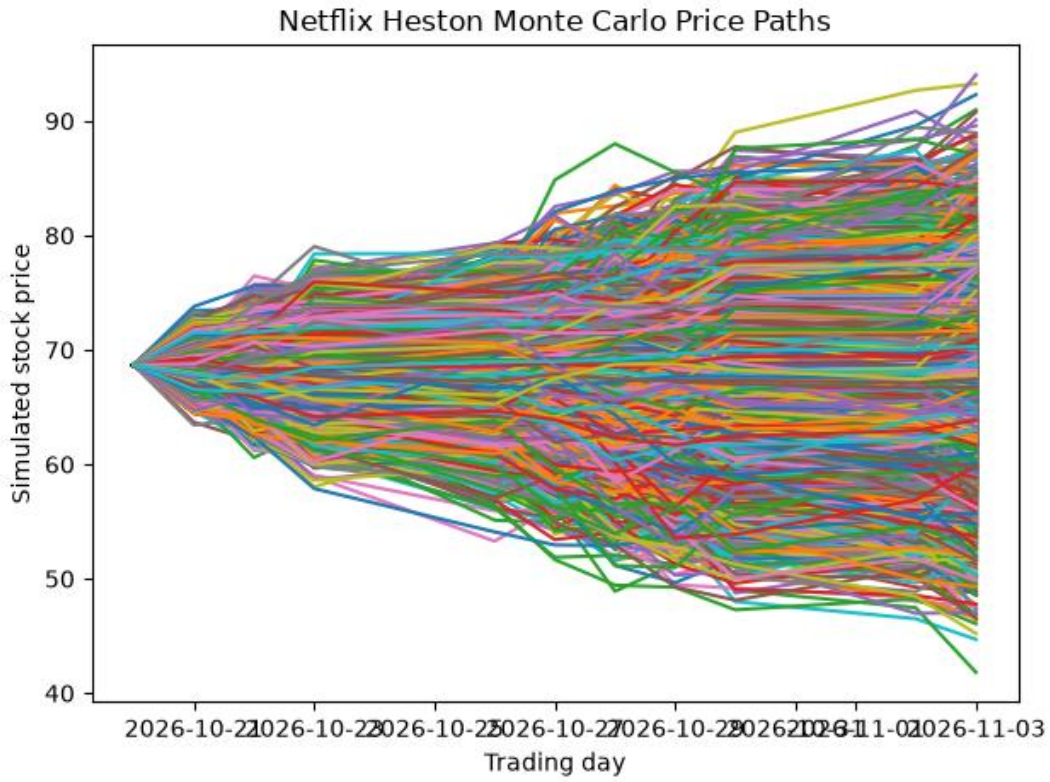
The simulation indicates a small positive average return and a probability of profit above 50%, but the downside tail remains material. The 95% Expected Shortfall is considerably larger than the 95% Value at Risk, showing that losses can deepen substantially once outcomes enter the worst 5% of the distribution.

1. Methodology

The model uses historical Netflix closing prices to calculate daily log returns and a 20-day annualized rolling variance. A simplified regression-based calibration estimates the Heston variance parameters, after which 50,000 correlated price-and-variance paths are simulated over 10 business days.

- Simulation start: 20 October 2026
- Simulation terminal date: 3 November 2026
- Simulation horizon: 10 business-day steps
- Number of simulated paths: 50,000
- Risk measures: 95% Value at Risk and 95% Expected Shortfall

2. Latest simulation results

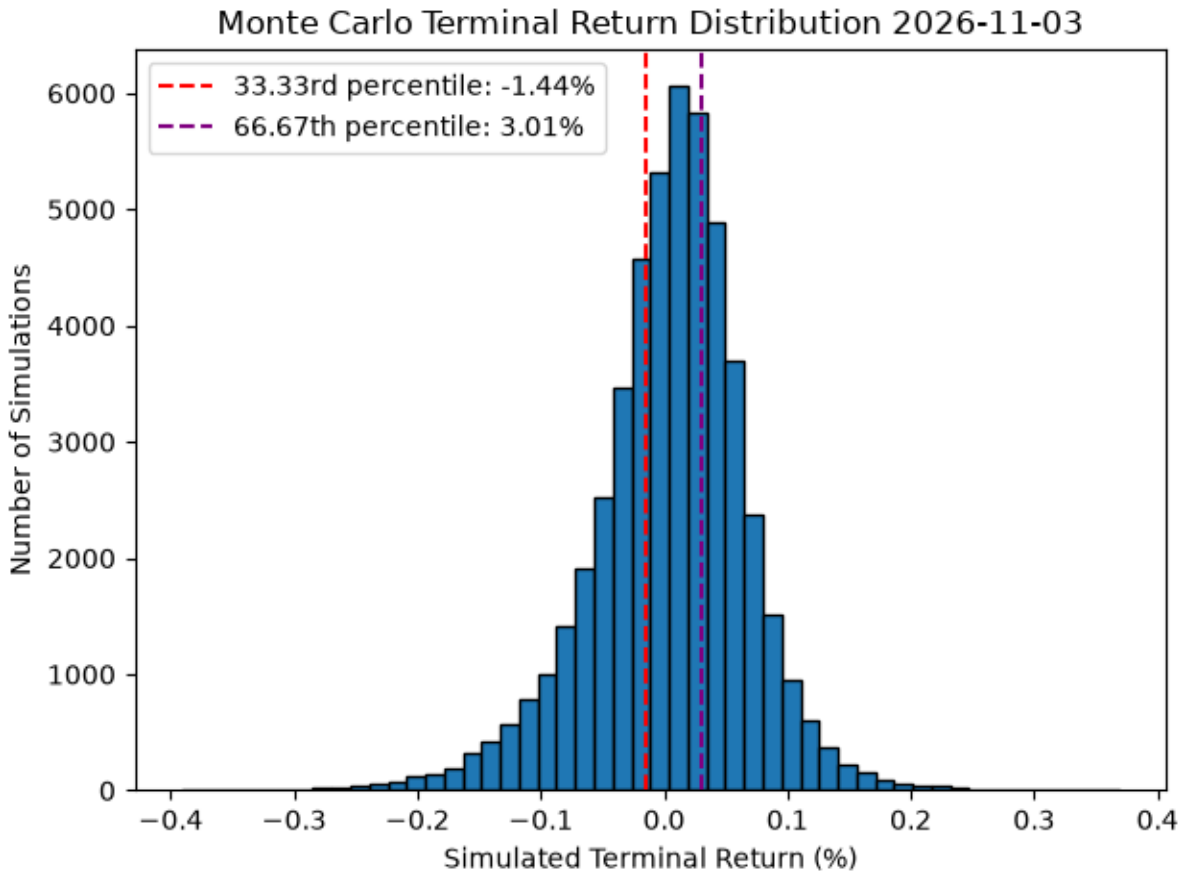


Measure	Result	Interpretation
Mean terminal return	**0.471%**	Average simulated holding-period return.
Median terminal return	**0.958%**	Half of the simulations end above this return and half below.
Median terminal price	**$69.348**	Central terminal-price outcome across all paths.
Probability of profit	**57.530%**	Share of simulations with terminal price above initial price.
95% Value at Risk	**10.802% loss**	Estimated loss threshold exceeded in about 5% of simulations.
95% Expected Shortfall	**15.308% loss**	Average loss among outcomes in the worst 5% tail.

3. Scenario analysis

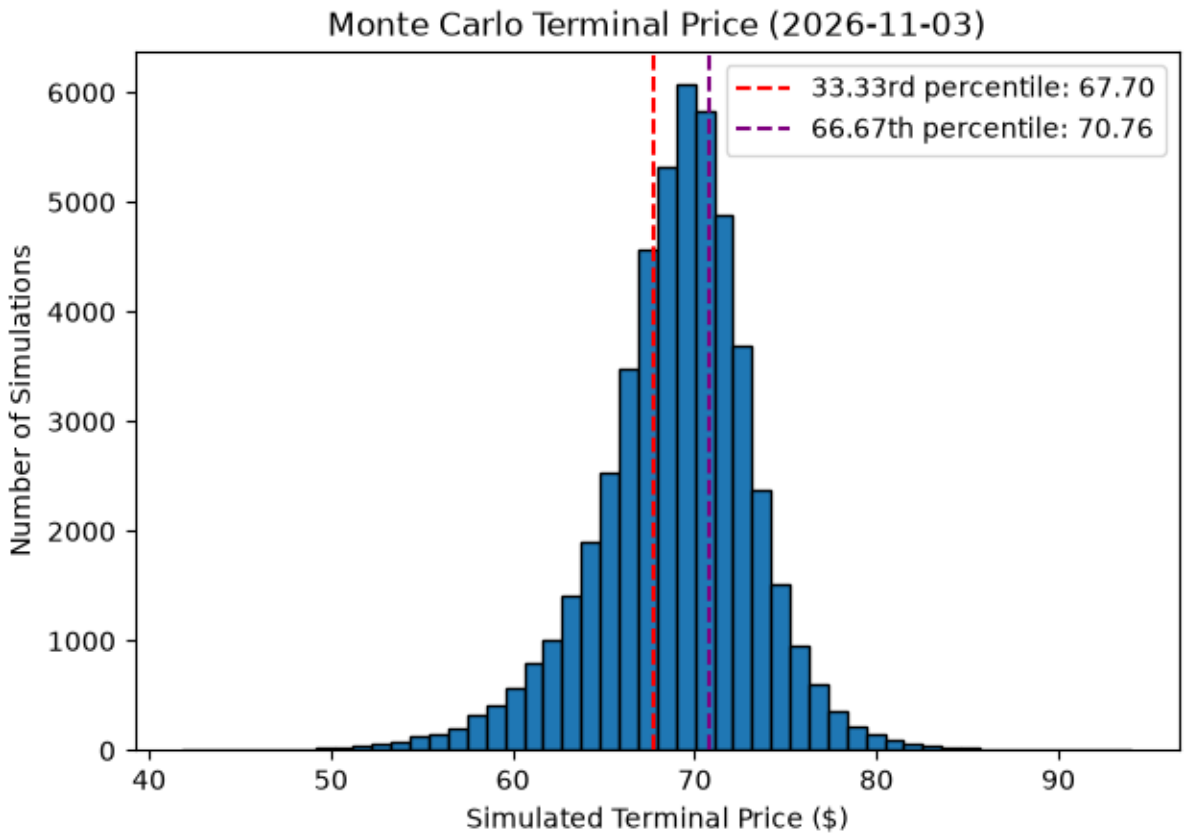
The 33rd and 67th percentiles divide the simulated outcomes into three approximately equal-probability groups. The middle-third interval is the central third of the distribution, not a range containing most outcomes.

Terminal return scenarios



Scenario	Terminal return
Lower third	Below -1.31%
Middle third	-1.31% to 3.07%
Upper third	Above 3.07%

Terminal price scenarios



Scenario	Terminal price
Lower third	Below $67.79
Middle third	$67.79 to $70.80
Upper third	Above $70.80

4. Interpretation

The simulated distribution has a positive central tendency: the mean return is 0.471%, the median return is 0.958%, and 57.530% of paths are profitable. The median exceeding the mean suggests that more adverse tail outcomes pull down the average return.

Downside risk remains important. A 95% VaR of 10.802% means the loss is greater than approximately 10.802% in the worst 5% of simulated outcomes. Conditional on entering that tail, the average loss rises to 15.308%, as measured by Expected Shortfall. Expected Shortfall therefore provides more information about the severity of extreme losses than VaR alone.

5. Conclusion

The 10-business-day Heston simulation indicates modest upside potential, with a 0.471% mean return and a 57.530% probability of profit. Nevertheless, the 15.308% Expected Shortfall shows that adverse tail outcomes

can be substantial. The results support cautious position sizing and explicit downside-risk controls rather than relying only on the positive average outcome.

Model limitations. The parameters are estimated from historical returns and rolling variance using a simplified regression approach rather than full maximum-likelihood or option-market calibration. The 10-business-day calendar excludes weekends but may not fully reflect US market holidays. Nasdaq closing-price data should also be checked for corporate-action adjustments. The results are model-generated scenarios, not guaranteed forecasts or investment recommendations.